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Acoustic wave filter with wide bandwidth of attenuation region

Abstract

An acoustic wave filter provided herein comprises a plurality of series resonators, and a plurality of shunt resonators disposed between the series resonators and a ground, at least one shunt resonator including a substrate, a pair of IDT electrodes disposed on the substrate, each of the IDT electrodes including a bus bar and a plurality of fingers extending from the bus bar, the plurality of fingers of the pair of IDT electrodes including a first group of fingers located at a center region of the pair of IDT electrodes and a second group of fingers located at edge regions on both sides of the center region, the first group of fingers having an average pitch distance shorter than an average pitch distance of the second group of fingers to improve a response at a rejection band of the acoustic wave filter.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application Ser. No. 63/307,684, titled “ACOUSTIC WAVE FILTER WITH WIDE BANDWIDTH OF ATTENUATION REGION,” filed Feb. 8, 2022, the entire content of which is incorporated herein by reference for all purposes.

BACKGROUND

Field

(1) Aspects and embodiments disclosed herein relate to electronic systems, and in particular, to a filter for use in radio frequency (RF) electronics.

Description of the Related Technology

(2) Filters are used in radio frequency (RF) communication systems to allow signals within specified frequency ranges to pass through and to reject frequencies outside of the specified ranges. An acoustic wave filter, which is used widely in the wireless communication field, can include a

plurality of resonators arranged to filter a radio frequency signal. Examples of acoustic wave filters include surface acoustic wave (SAW) filters and bulk acoustic wave (BAW) filters. A film bulk acoustic resonator (FBAR) filter is an example of a BAW filter. Acoustic wave filters can be implemented in radio frequency electronic systems. For instance, filters in a radio frequency front end of a mobile phone can include acoustic wave filters. A plurality of acoustic wave filters can be arranged as a multiplexer. For example, two surface acoustic wave filters can be arranged as a duplexer.

(3) Examples of RF communication systems with one or more filter module include, but are not limited to, mobile phones, tablets, base stations, network access points, customer-premises equipment (CPE), laptops, and wearable electronics. For example, in wireless devices that communicate using a cellular standard, a wireless local area network (WLAN) standard, and/or any other suitable communication standard, a power amplifier can be used for RF signal amplification. An RF signal can have a frequency in the range of about 30 kHz to 300 GHz, such as in the range of about 410 MHz to about 7.125 GHz for certain communications standards.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram of one embodiment of a mobile device;
- (2) FIG. 2A is a schematic diagram of a carrier aggregation system;
- (3) FIG. 2B is a schematic diagram of a carrier aggregation system;
- (4) FIG. 2C is a schematic diagram of a carrier aggregation system;
- (5) FIG. 2D is a schematic diagram of a carrier aggregation system;
- (6) FIG. 3A is a schematic block diagram of a module that includes a filter;
- (7) FIG. 3B is a schematic block diagram of a module that includes a filter;
- (8) FIG. 4 is a schematic diagram of an example of an acoustic wave device;
- (9) FIG. 5 is a sectional view of the acoustic wave device aligned along a single electrode finger;
- (10) FIG. 6 is a schematic diagram of an acoustic wave filter according to an embodiment;
- (11) FIG. 7A is a schematic diagram illustrating pitch distance of each part of a resonator according to the prior art;
- (12) FIG. 7B illustrates average pitch distances of fingers depending on their location;
- (13) FIG. 8 is an example of passband characteristics of an acoustic wave filter according to the prior art;
- (14) FIG. 9A is a schematic diagram illustrating pitch distance of each part of a resonator according to an embodiment;
- (15) FIG. 9B illustrates average pitch distances of fingers depending on their location according to an embodiment;
- (16) FIG. 10A is an example of the frequency response of a resonator according to an embodiment;
- (17) FIG. 10B is an example of the passband characteristics of an acoustic wave filter according to an embodiment;
- (18) FIG. 11 is an example of the passband characteristics of an acoustic wave filter according to an embodiment;
- (19) FIG. 12A is a schematic diagram of one embodiment of a packaged module;
- (20) FIG. 12B is a cross-section of the packaged module of FIG. 12A taken along the lines 12B-12B; and
- (21) FIG. 13 is a schematic diagram of one embodiment of a phone board.

DETAILED DESCRIPTION OF EMBODIMENTS

(22) The following detailed description of certain embodiments presents various descriptions of specific embodiments. However, the innovations described herein can be embodied in a multitude

of different ways, for example, as defined and covered by the claims. In this description, reference is made to the drawings where like reference numerals can indicate identical or functionally similar elements. It will be understood that elements illustrated in the figures are not necessarily drawn to scale. Moreover, it will be understood that certain embodiments can include more elements than illustrated in a drawing and/or a subset of the elements illustrated in a drawing. Further, some embodiments can incorporate any suitable combination of features from two or more drawings.

(23) FIG. 1 is a schematic diagram of one example of a mobile device **100**. The mobile device **100** includes a baseband system **101**, a transceiver **102**, a front end system **103**, antennas **104**, a power management system **105**, a memory **106**, a user interface **107**, and a battery **108**.

(24) The mobile device **100** can be used communicate using a wide variety of communications technologies, including, but not limited to, 2G, 3G, 4G (including LTE, LTE-Advanced, and LTE-Advanced Pro), 5G, WLAN (for instance, Wi-Fi), WPAN (for instance, Bluetooth and ZigBee), WMAN (for instance, WiMax), and/or GPS technologies.

(25) The transceiver **102** generates RF signals for transmission and processes incoming RF signals received from the antennas **104**. It will be understood that various functionalities associated with the transmission and receiving of RF signals can be achieved by one or more components that are collectively represented in FIG. 1 as the transceiver **102**. In one example, separate components (for instance, separate circuits or dies) can be provided for handling certain types of RF signals.

(26) The front end system **103** aids in conditioning signals transmitted to and/or received from the antennas **104**. In the illustrated embodiment, the front end system **103** includes power amplifiers (PAs) **111**, low noise amplifiers (LNAs) **112**, filters **113**, switches **114**, and duplexers **115**. However, other implementations are possible.

(27) For example, the front end system **103** can provide a number of functionalities, including, but not limited to, amplifying signals for transmission, amplifying received signals, filtering signals, switching between different bands, switching between different power modes, switching between transmission and receiving modes, duplexing of signals, multiplexing of signals (for instance, diplexing or triplexing), or some combination thereof.

(28) In certain implementations, the mobile device **100** supports carrier aggregation, thereby providing flexibility to increase peak data rates. Carrier aggregation can be used for both Frequency Division Duplexing (FDD) and Time Division Duplexing (TDD), and may be used to aggregate a plurality of carriers or channels. Carrier aggregation includes contiguous aggregation, in which contiguous carriers within the same operating frequency band are aggregated. Carrier aggregation can also be non-contiguous, and can include carriers separated in frequency within a common band and/or in different bands.

(29) The antennas **104** can include antennas used for a wide variety of types of communications. For example, the antennas **104** can include antennas associated with transmitting and/or receiving signals associated with a wide variety of frequencies and communications standards.

(30) In certain implementations, the antennas **104** support multiple-input-multiple-output (MIMO) communications and/or switched diversity communications. For example, MIMO communications use multiple antennas for communicating multiple data streams over a single radio frequency channel. MIMO communications benefit from higher signal to noise ratio, improved coding, and/or reduced signal interference due to spatial multiplexing differences of the radio environment. Switched diversity refers to communications in which a particular antenna is selected for operation at a particular time. For example, a switch can be used to select a particular antenna from a group of antennas based on a variety of factors, such as an observed bit error rate and/or a signal strength indicator.

(31) The mobile device **100** can operate with beamforming in certain implementations. For example, the front end system **103** can include phase shifters having variable phases controlled by the transceiver **102**. Additionally, the phase shifters are controlled to provide beam formation and directivity for transmission and/or reception of signals using the antennas **104**. For example, in the

context of signal transmission, the phases of the transmit signals provided to the antennas **104** are controlled such that radiated signals from the antennas **104** combine using constructive and destructive interference to generate an aggregate transmit signal exhibiting beam-like qualities with more signal strength propagating in a given direction. In the context of signal reception, the phases are controlled such that more signal energy is received when the signal is arriving at the antennas **104** from a particular direction. In certain implementations, the antennas **104** include one or more arrays of antenna elements to enhance beamforming.

(32) The baseband system **101** is coupled to the user interface **107** to facilitate processing of various user input and output (I/O), such as voice and data. The baseband system **101** provides the transceiver **102** with digital representations of transmit signals, which the transceiver **102** processes to generate RF signals for transmission. The baseband system **101** also processes digital representations of received signals provided by the transceiver **102**. As shown in FIG. 1, the baseband system **101** is coupled to the memory **106** to facilitate operation of the mobile device **100**.

(33) The memory **106** can be used for a wide variety of purposes, such as storing data and/or instructions to facilitate the operation of the mobile device **100** and/or to provide storage of user information.

(34) The power management system **105** provides a number of power management functions of the mobile device **100**. The power management system **105** of FIG. 1 includes an envelope tracker **160**. As shown in FIG. 1, the power management system **105** receives a battery voltage form the battery **108**. The battery **108** can be any suitable battery for use in the mobile device **100**, including, for example, a lithium-ion battery.

(35) The mobile device **100** of FIG. 1 illustrates one example of an RF communication system that can include power amplifier(s) implemented in accordance with one or more features of the present disclosure. However, the teachings herein are applicable to RF communication systems implemented in a wide variety of ways.

(36) FIG. 2A is a schematic diagram of a carrier aggregation system **40**. The illustrated carrier aggregation system **40** includes power amplifiers **42A** and **42B**, switches **43A** and **43B**, duplexers **44A** and **44B**, switches **45A** and **45B**, diplexer **46**, and antenna **47**. The power amplifiers **42A** and **42B** can each transmit an amplified RF signal associated with a different carrier. The switch **43A** can be a band select switch. The switch **43A** can couple an output of the power amplifier **42A** to a selected duplexer of the duplexers **44A**. Each of the duplexers can include a transmit filter and receive filter. Any of the filters of the duplexers **44A** and **44B** can be implemented in accordance with any suitable principles and advantages discussed herein. The switch **45A** can couple the selected duplexer of the duplexers **44A** to the diplexer **46**. The diplexer **46** can combine RF signals provided by the switches **45A** and **45B** into a carrier aggregation signal that is transmitted by the antenna **47**. The diplexer **46** can isolate different frequency bands of a carrier aggregation signal received by the antenna **47**. The diplexers **46** is an example of a frequency domain multiplexer. Other frequency domain multiplexers include a triplexer. Carrier aggregation systems that include triplexers can process carrier aggregation signals associated with three carriers. The switches **45A** and **45B** and selected receive filters of the duplexers **44A** and **44B** can provide RF signals with the isolated frequency bands to respective receive paths.

(37) FIG. 2B is a schematic diagram of a carrier aggregation system **50**. The illustrated carrier aggregation system **50** includes power amplifiers **42A** and **42B**, low noise amplifiers **52A** and **52B**, switches **53A** and **53B**, filters **54A** and **54B**, diplexer **46**, and antenna **47**. The power amplifiers **42A** and **42B** can each transmit an amplified RF signal associated with a different carrier. The switch **53A** can be a transmit/receive switch. The switch **53A** can couple the filter **54A** to an output of the power amplifier **42A** in a transmit mode and to an input of the low noise amplifier **52A** in a receive mode. The filter **54A** and/or the filter **54B** can be implemented in accordance with any suitable principles and advantages discussed herein. The diplexer **46** can combine RF signals from the power amplifiers **42A** and **42B** provided by the switches **53A** and **53B** into a carrier aggregation

signal that is transmitted by the antenna **47**. The diplexer **46** can isolate different frequency bands of a carrier aggregation signal received by the antenna **47**. The switches **53A** and **53B** and the filters **54A** and **54B** can provide RF signals with the isolated frequency bands to respective low noise amplifiers **52A** and **52B**.

(38) FIG. 2C is a schematic diagram of a carrier aggregation system **60** that includes multiplexers in signal paths between power amplifiers and an antenna. The illustrated carrier aggregation system **60** includes a low band path, a medium band path, and a high band path. In certain implementations, a low band path can process radio frequency signals having a frequency of less than 1 GHz, a medium band path can process radio frequency signals having a frequency between 1 GHz and 2.2 GHz, and a high band path can process radio frequency signals having a frequency above 2.2 GHz.

(39) A diplexer **46** can be included between RF signal paths and an antenna **47**. The diplexer **46** can frequency multiplex radio frequency signals that are relatively far away in frequency. The diplexer **46** can be implemented with passive circuit elements having a relatively low loss. The diplexer **46** can combine (for transmit) and separate (for receive) carriers of carrier aggregation signals.

(40) As illustrated, the low band path includes a power amplifier **42A** configured to amplify a low band radio frequency signal, a band select switch **43A**, and a multiplexer **64A**. The band select switch **43A** can electrically connect the output of the power amplifier **42A** to a selected transmit filter of the multiplexer **64A**. The selected transmit filter can be a band pass filter with a pass band corresponding to a frequency of an output signal of the power amplifier **42A**. The multiplexer **64A** can include any suitable number of transmit filters and any suitable number of receive filters. One or more of the transmit filters and/or one or more of the receive filters can be implemented in accordance with any suitable principles and advantages discussed herein. The multiplexer **64A** can have the same number of transmit filters as receive filters. In some instances, the multiplexer **64A** can have a different number of transmit filters than receive filters.

(41) As illustrated in FIG. 2C, the medium band path includes a power amplifier **42B** configured to amplify a medium band radio frequency signal, a band select switch **43B**, and a multiplexer **64B**. The band select switch **43B** can electrically connect the output of the power amplifier **42B** to a selected transmit filter of the multiplexer **64B**. The selected transmit filter can be a band pass filter with a pass band corresponding to a frequency of an output signal of the power amplifier **42B**. The multiplexer **64B** can include any suitable number of transmit filters and any suitable number of receive filters. One or more of the transmit filters and/or one or more of the receive filters can be implemented in accordance with any suitable principles and advantages discussed herein. The multiplexer **64B** can have the same number of transmit filters as receive filters. In some instances, the multiplexer **64B** can have a different number of transmit filters than receive filters.

(42) In the illustrated carrier aggregation system **60**, the high band path includes a power amplifier **42C** configured to amplify a high band radio frequency signal, a band select switch **43C**, and a multiplexer **64C**. The band select switch **43C** can electrically connect the output of the power amplifier **42C** to a selected transmit filter of the multiplexer **64C**. The selected transmit filter can be a band pass filter with a pass band corresponding to a frequency of an output signal of the power amplifier **42C**. The multiplexer **64C** can include any suitable number of transmit filters and any suitable number of receive filters. One or more of the transmit filters and/or one or more of the receive filters can be implemented in accordance with any suitable principles and advantages discussed herein. The multiplexer **64C** can have the same number of transmit filters as receive filters. In some instances, the multiplexer **64C** can have a different number of transmit filters than receive filters.

(43) A select switch **65** can selectively provide a radio frequency signal from the medium band path or the high band path to the diplexer **46**. Accordingly, the carrier aggregation system **60** can process carrier aggregation signals with either a low band and high band combination or a low band and medium band combination.

- (44) FIG. 2D is a schematic diagram of a carrier aggregation system **70** that includes multiplexers in signal paths between power amplifiers and an antenna. The carrier aggregation system **70** is like the carrier aggregation system **60** of FIG. 2C, except that the carrier aggregation system **70** includes switch-plexing features. Switch-plexing can be implemented in accordance with any suitable principles and advantages discussed herein.
- (45) Switch-plexing can implement on-demand multiplexing. Some radio frequency systems can operate in a single carrier mode for a majority of time (e.g., about 95% of the time) and in a carrier aggregation mode for a minority of the time (e.g., about 5% of the time). Switch-plexing can reduce loading in a single carrier mode in which the radio frequency system can operate for the majority of the time relative to a multiplexer that includes filters having a fixed connection at a common node. Such a reduction in loading can be more significant when there are a relatively larger number of filters included in multiplexer.
- (46) In the illustrated carrier aggregation system **70**, duplexers **44B** and **44C** are selectively coupled to a diplexer **46** by way of a switch **75**. The switch **75** is configured as a multi-close switch that can have two or more throws active concurrently. Having multiple throws of the switch **75** active concurrently can enable transmission and/or reception of carrier aggregation signals. The switch **75** can also have a single throw active during a single carrier mode. As illustrated, each duplexer of the duplexers **44A**, **44B** is coupled to separate throws of the switch **75**. Similarly, the illustrated duplexers **44A**, **44B** include a plurality of duplexers coupled to separate throws of the switch **75**. Alternatively, instead of duplexers being coupled to each throw the switch **75** as illustrated in FIG. 2D, one or more individual filters of a multiplexer can be coupled to a dedicated throw of a switch coupled between the multiplexer and a common node. For instance, in some implementations, such a switch could have twice as many throws as the illustrated switch **75**.
- (47) The filters discussed herein can be implemented in a variety of packaged modules. Some examples of packaged modules will now be discussed in which any suitable principles and advantages of the filters discussed herein can be implemented. FIGS. 3A and 3B are schematic block diagrams of illustrative packaged modules according to certain embodiments.
- (48) FIG. 3A is a schematic block diagram of a module **80** that includes a power amplifier **42**, a switch **83**, and filters **84** in accordance with one or more embodiments. The module **80** can include a package that encloses the illustrated elements. The power amplifier **42**, a switch **83**, and filters **84** can be disposed on the same packaging substrate. The packaging substrate can be a laminate substrate, for example. The switch **83** can be a multi-throw radio frequency switch. The switch **83** can electrically couple an output of the power amplifier **42** to a selected filter of the filters **84**. The filters **84** can include any suitable number of surface acoustic wave filters. One or more filters of the filters **84** can be implemented in accordance with any suitable principles and advantages disclosed herein.
- (49) FIG. 3B is a schematic block diagram of a module **85** that includes power amplifiers **42A** and **42B**, switches **83A** and **83B**, and filters **84A** and **84B** in accordance with one or more embodiments, and an antenna switch **88**. The module **85** is like the module **80** of FIG. 3A, except the module **85** includes an additional RF signal path and the antenna switch **88** arranged to selectively couple a signal from the filters **84A** or the filters **84B** to an antenna node. One or more filters of the filters **84A** and/or **84B** can be implemented in accordance with any suitable principles and advantages disclosed herein. The additional RF signal path includes an additional power amplifier **42B**, and additional switch **83B**, and additional filters **84B**. The different RF signal paths can be associated with different frequency bands and/or different modes of operation (e.g., different power modes, different signaling modes, etc.).
- (50) In recent years, in the field of information communication devices such as mobile phones, acoustic wave devices having a comb-shaped IDT electrode formed on a surface of a piezoelectric substrate have been used as circuit elements such as resonators, filters, and the like.
- (51) FIG. 4 shows an example of such an acoustic wave device. In FIG. 4, a top view of an acoustic

wave device **400** is shown. In this description, the acoustic wave device **400** can be also referred to as a resonator. The acoustic wave device **400** is formed by arranging two IDT electrodes **402** and two reflectors **403** on a piezoelectric substrate **401**. The IDT electrodes **402** each have a bus bar **411** and a plurality of electrode fingers **412** that extend from the bus bar **411**. The respective electrode fingers **412** of each of the IDT electrodes **402** are arranged such that the electrode fingers **412** thereof are arranged interleaved with the respective electrode fingers **412** of the other IDT electrode **402**. The reflectors **403** are arranged to interpose the IDT electrodes **402** therebetween.

(52) FIG. 5 is a sectional view of an acoustic wave device **500** similar to that of FIG. 4 aligned along a single electrode finger **412**. In the acoustic wave device **500**, propagation of an acoustic wave is concentrated by the coating film **504**, thereby suppressing unnecessary high-order transverse mode waves.

(53) FIG. 6 is a schematic diagram of an acoustic wave filter **600** according to an embodiment. In this embodiment, the acoustic wave filter **600** is a ladder type filter. The ladder type filter includes a plurality of series resonators (Rs1, Rs2, Rs3) and a plurality of shunt resonators (Rp1, Rp2, Rp3) disposed between nodes between adjacent series resonators (Rs1, Rs2, Rs3) and ground. Each of the plurality of series resonators and shunt resonators of the acoustic wave filter **600** can be implemented by an acoustic wave device as shown in FIG. 4.

(54) The electrical characteristics of the series and shunt resonators can be determined based on the required performance of the acoustic wave filter **600**. For example, the resonant frequency of the shunt resonators can be determined based on the targeted frequency range of the rejection band of the acoustic wave filter **600**.

(55) FIG. 7A is schematic diagram illustrating a pitch distance of fingers in each part of an example of a resonator **700** according to the prior art. The resonator **700** can be used as one of the plurality of series resonators or one of the plurality of shunt resonators of an acoustic wave filter such as acoustic wave filter **600**.

(56) The resonator **700** includes a substrate **702**, a pair of IDT electrodes **704**, and a dielectric film **708**. The resonator **700** may further include a pair of reflectors **706**. The substrate **702** is a piezoelectric substrate. The substrate **702** may be formed of lithium niobate (LiNbO₃). The pair of IDT electrodes **704** are configured to excite a main wave on the substrate **702**. Each of IDT electrodes **704** can be formed of at least one of aluminum (Al), copper (Cu), aluminum-magnesium-copper (AlMgCu) alloy, tungsten (W), platinum (Pt), and/or molybdenum (Mo). The dielectric film **708** is configured to cover at least a part of the substrate **702** and the pair of IDT electrodes **704**. The dielectric film **708** is formed of silicon dioxide (SiO₂).

(57) Each of the pair of IDT electrodes **704** includes a bus bar extending along a propagating direction x of a main acoustic wave, and a plurality of fingers extending from the bus bar, as illustrated in FIG. 4. The extending direction of the plurality of fingers is perpendicular to the length direction of the bus bar. That is, the plurality of fingers extend in a perpendicular direction to the propagating direction x of the main acoustic wave.

(58) The bus bars of the pair of IDT electrodes **704** are arranged in parallel to each other. The fingers of one IDT electrode extend toward the bus bar of the other IDT electrode. The respective fingers of one IDT electrode are arranged such that the fingers thereof are arranged interleaved with the respective fingers of the other IDT electrode. A pitch of the fingers of one IDT electrode indicates a distance between the respective finger and an adjacent finger of the other IDT electrode.

(59) The pair of IDT electrodes **704** has a center region **704-1** and edge regions **704-2** on both sides of the center region **704-1**. The width of the center region **704-1** can be determined based on a desired performance of the acoustic wave filter in which the resonator **700** is used. The edge regions **704-2** of the pair of IDT electrodes **704** are positioned before and after the center region **704-1** along the propagating direction x of the main acoustic wave.

(60) The reflectors **706** interpose the pair of IDT electrodes **704** therebetween. The edge regions **704-2** of the pair of IDT electrodes **704** are positioned on both sides of the center region **704-1** that

are adjacent to the reflectors **706**. Each of the reflectors **706** includes a pair of bus bars arranged in parallel to each other, and a plurality of fingers connecting the pair of bus bars.

(61) According to the prior art, an average pitch distance ($\lambda_{\text{sub.1}}$) of fingers of the center region **704-1** of the pair of IDT electrodes **704** is equal to or longer than an average pitch distance ($\lambda_{\text{sub.2}}$, $\lambda_{\text{sub.3}}$) of fingers of the edge regions **704-2**. In addition, an average pitch distance ($\lambda_{\text{sub.4}}$, $\lambda_{\text{sub.5}}$) of the fingers of the reflectors **706** is longer than the average pitch distance ($\lambda_{\text{sub.2}}$, $\lambda_{\text{sub.3}}$) of the fingers of the edge regions **704-2**.

(62) FIG. 7B is example of average pitch distances of fingers of the resonator of FIG. 7A depending on their location. As shown in FIG. 7B, an average pitch distance ($\lambda_{\text{sub.1}}$) of the fingers of the center region **704-1** of the pair of IDT electrodes **704** is equal to or longer than an average pitch distance ($\lambda_{\text{sub.2}}$, $\lambda_{\text{sub.3}}$) of the fingers of the edge regions **704-2**. In addition, an average pitch distance ($\lambda_{\text{sub.4}}$, $\lambda_{\text{sub.5}}$) of the fingers of the reflectors **706** is longer than the average pitch distance ($\lambda_{\text{sub.2}}$, $\lambda_{\text{sub.3}}$) of the fingers of the edge regions **704-2**.

(63) FIG. 8 is an example of passband characteristics of an example of an acoustic wave filter according to the prior art. In this example, the acoustic wave filter can be a ladder type filter as shown in FIG. 6. The acoustic wave filter includes a plurality of series resonators (Rs1, Rs2, Rs3) and a plurality of shunt resonators (Rp1, Rp2, Rp3).

(64) As shown in FIG. 8, the acoustic wave filter has a pass band and rejection bands on both sides of the passband. The frequency ranges of the rejection bands of the acoustic wave filter are determined based on the characteristics of the shunt resonators (Rs1, Rs2, Rs3). In particular, the rejection band on the lower frequency side of the passband is affected by the attenuation pole of each of the shunt resonators (Rp1, Rp2, Rp3) that corresponds to the resonant frequency of individual shunt resonators (Rp1, Rp2, Rp3).

(65) As the rejection bands of the acoustic wave filter are broadened, the characteristics of the acoustic wave filter can be improved. Therefore, it is desired to enhance the bandwidth of the rejection bands of the acoustic wave filter such to improve the overall performance of the acoustic wave filter.

(66) Hereinafter, an acoustic wave filter with wide rejection bands according to an embodiment is provided.

(67) FIG. 9A is a schematic diagram illustrating pitch distance of fingers in each part of a resonator **900** according to an embodiment. The resonator **900** can be used as one of the plurality of series resonators or one of the plurality of shunt resonators of the acoustic wave filter **600** as shown in FIG. 6.

(68) The resonator **900** includes a substrate **902**, a pair of IDT electrodes **904**, and a dielectric film **908**. The resonator **900** may further include a pair of reflectors **906**. The substrate **902** is a piezoelectric substrate. The substrate **902** may be formed of lithium niobate (LiNbO_{sub.3}). The pair of IDT electrodes **904** are configured to excite a main acoustic wave on the substrate **902**. Each IDT electrode **904** can be formed of at least one of aluminum (Al), copper (Cu), aluminum-magnesium-copper (AlMgCu) alloy, tungsten (W), platinum (Pt), and/or molybdenum (Mo). The dielectric film **908** covers at least a part of the substrate **902** and the pair of IDT electrodes **904**. The dielectric film **908** is formed of silicon dioxide (SiO_{sub.2}).

(69) Each IDT electrode **904** includes a bus bar extending along a propagating direction x of a main acoustic wave, and a plurality of fingers extending from the bus bar, as illustrated in FIG. 4. The extending direction of the plurality of fingers is perpendicular to the length direction of the bus bar. That is, the plurality of fingers extend in a perpendicular direction to the propagating direction x of the main acoustic wave.

(70) The bus bars of the pair of IDT electrodes **904** are arranged in parallel to each other. The fingers of one IDT electrode extend toward the bus bar of the other IDT electrode. The respective fingers of one IDT electrode are interleaved with the respective fingers of the other IDT electrode. A pitch of a respective finger of one IDT electrode indicates a distance between the respective

finger and an adjacent finger of the other IDT electrode.

(71) The pair of IDT electrodes **904** has a center region **904-1** and edge regions **904-2** on both sides of the center region **904-1**. The width of the center region **904-1** can be determined based on a desired performance of an acoustic wave filter in which the resonator **900** is used. The edge regions **904-2** of the pair of IDT electrodes **904** are positioned before and after the center region **904-1** along the propagating direction x of a main acoustic wave.

(72) The plurality of fingers of the pair of IDT electrodes **904** include a first group of fingers and a second group of fingers. The first group of fingers is located in the center region **904-1**, and the second group of fingers is located in the edge regions **904-2**. At least a part of second group of fingers is located on a first side of the center region **904-1** and another part of the second group of fingers is located on a second side of the center region **904-1**. The first group of fingers consists of a part of one IDT electrode and a part of the other IDT electrode. The second group of fingers consists of another part of one IDT electrode and another part of the other IDT electrode.

(73) The reflectors **906** interpose the pair of IDT electrodes **904** therebetween. The edge regions **904-2** of the pair of IDT electrodes **904** are positioned on both sides of the center region **904-1** that are adjacent to the reflectors **906**. Each of the reflectors **906** includes a pair of bus bars arranged in parallel to each other, and a plurality of fingers connecting the pair of bus bars.

(74) According to an aspect of the present disclosure, an average pitch distance ($\lambda_{\text{sub.1}}$) of the fingers of the center region **904-1** of the pair of IDT electrodes **904** (the first group of fingers) is shorter than an average pitch distance ($\lambda_{\text{sub.2}}$, $\lambda_{\text{sub.3}}$) of the fingers of the edge regions **904-2** (the second group of fingers). The pair of IDT electrodes **904** is configured to induce a longitudinal mode response. The longitudinal mode response is formed by waves confined in a resonant cavity of the acoustic wave filter. The longitudinal mode of a resonant cavity is a particular standing wave pattern formed by waves confined in the cavity. In this manner, the bandwidths of the rejection bands of the acoustic wave filter may be increased. Each finger of the first group of fingers has different pitch distance with adjacent fingers. The pitch distance of each finger can be determined based on the desired performance of the acoustic wave filter, such as frequency range of the passband. By intentionally inducing the longitudinal mode response in at least one shunt resonator, the bandwidths of the rejection bands of the acoustic wave filter can be increased.

(75) In addition, an average pitch distance ($\lambda_{\text{sub.4}}$, $\lambda_{\text{sub.5}}$) of the fingers of the reflectors **906** are longer than the average pitch distance ($\lambda_{\text{sub.2}}$, $\lambda_{\text{sub.3}}$) of the fingers of the edge regions **904-2**.

(76) FIG. **9B** is example of average pitch distances of fingers depending on their location according to an embodiment. As shown in FIG. **9B**, the average pitch distance ($\lambda_{\text{sub.1}}$) of fingers of the center region **904-1** of the pair of IDT electrodes **904** (the first group of fingers) is shorter than the average pitch distance ($\lambda_{\text{sub.2}}$, $\lambda_{\text{sub.3}}$) of the fingers of the edge regions **904-2** (the second group of fingers). In addition, the average pitch distance ($\lambda_{\text{sub.4}}$, $\lambda_{\text{sub.5}}$) of the fingers of the reflectors **906** is longer than the average pitch distance ($\lambda_{\text{sub.2}}$, $\lambda_{\text{sub.3}}$) of the fingers of the edge regions **904-2**.

(77) FIG. **10A** is an example of frequency response of a resonator according to an embodiment. The Y-axis of FIG. **10A** indicates conductance [dB] of the resonator and the X-axis indicates frequency. As shown in FIG. **10A**, the response of the resonator has many peaks in conductance values, which represent a longitudinal mode response.

(78) FIG. **10B** is an example of passband characteristics of an acoustic wave filter according to an embodiment. In this embodiment, the acoustic wave filter can be a ladder type filter as shown in FIG. **6**. The acoustic wave filter includes a plurality of series resonators (R_{s1} , R_{s2} , R_{s3}) and a plurality of shunt resonators ($R_{p1'}$, $R_{p2'}$, $R_{p3'}$).

(79) As shown in FIG. **10B**, the acoustic wave filter according to the embodiment has a pass band and rejection bands on both sides of the passband. The rejection bands of the acoustic wave filter are determined based on the frequency response of each of the shunt resonators ($R_{p1'}$, $R_{p2'}$, $R_{p3'}$). In particular, the rejection band in the frequency range below the passband is affected by the

longitudinal mode response from each of the shunt resonators. In other words, the resonant frequencies of the shunt resonators (Rp1', Rp2', Rp3') are located in the rejection band of the acoustic wave filter, and the longitudinal mode responses that appear in vicinity of the resonant frequencies of each shunt resonator (Rp1', Rp2', Rp3') affects the characteristics of the rejection band of the acoustic wave filter.

(80) FIG. 11 is an example of passband characteristics of an acoustic wave filter according to an embodiment. In this embodiment, the acoustic wave filter can be a ladder type filter as shown in FIG. 6. The acoustic wave filter includes a plurality of series resonators (Rs1, Rs2, Rs3) and a plurality of shunt resonators (Rp1', Rp2', Rp3').

(81) As shown in FIG. 11, the acoustic wave filter according to an embodiment has broader rejection band (curve (b)) in the frequency range below the passband as compared to the rejection band (curve (a)) of the acoustic wave filter according to the prior art. Since the shunt resonators (Rp1', Rp2', Rp3') according to this embodiment are configured to induce the longitudinal mode response, the rejection band of the acoustic wave filter does not have attenuation poles. Therefore, the bandwidth of the rejection band of the acoustic wave filter according to this embodiment can be increased.

(82) FIG. 12A is a schematic diagram of one embodiment of a packaged module 800. FIG. 12B is a schematic diagram of a cross-section of the packaged module 800 of FIG. 12A taken along the lines 12B-12B.

(83) The packaged module 800 includes an IC or die 801, surface mount components 803, wirebonds 808, a package substrate 820, and encapsulation structure 840. The package substrate 820 includes pads 806 formed from conductors disposed therein. Additionally, the die 801 includes pads 804, and the wirebonds 808 electrically connect the pads 804 of the die 801 to the pads 806 of the package substrate 820.

(84) The die 801 includes a filter module and a power amplifier 846, which can be implemented in accordance with any of the embodiments herein.

(85) The packaging substrate 820 can be configured to receive a plurality of components such as the die 801 and the surface mount components 803, which can include, for example, surface mount capacitors and/or inductors.

(86) As shown in FIG. 12B, the packaged module 800 includes a plurality of contact pads 832 disposed on the side of the packaged module 800 opposite the side used to mount the die 801. Configuring the packaged module 800 in this manner can aid in connecting the packaged module 800 to a circuit board such as a phone board of a wireless device. The contact pads 832 can be configured to provide RF signals, bias signals, power low voltage(s) and/or power high voltage(s) to the die 801 and/or the surface mount components 803. As shown in FIG. 12B, the electrically connections between the contact pads 832 and the die 801 can be facilitated by connections 833 through the package substrate 820. The connections 833 can represent electrical paths formed through the package substrate 820, such as connections associated with vias and conductors of a multilayer laminated package substrate.

(87) In some embodiments, the packaged module 800 can also include one or more packaging structures to, for example, provide protection and/or facilitate handling of the packaged module 800. Such a packaging structure can include overmold or encapsulation structure 840 formed over the packaging substrate 820 and the components and die(s) disposed thereon.

(88) It will be understood that although the packaged module 800 is described in the context of electrical connections based on wirebonds, one or more features of the present disclosure can also be implemented in other packaging configurations, including, for example, flip-chip configurations.

(89) FIG. 13 is a schematic diagram of one embodiment of a phone board 900. The phone board 900 includes the module 800 shown in FIGS. 12A-12B attached thereto. Although not illustrated in FIG. 13 for clarity, the phone board 900 can include additional components and structures.

(90) Some of the embodiments described above have provided examples in connection with

wireless devices or mobile phones. However, the principles and advantages of the embodiments can be used for any other systems or apparatus that have needs for power amplifiers.

(91) Embodiments as disclosed herein can be implemented in various electronic devices. Examples of the electronic devices can include, but are not limited to, consumer electronic products, parts of the consumer electronic products, electronic test equipment, etc. Examples of the electronic devices can also include, but are not limited to, memory chips, memory modules, circuits of optical networks or other communication networks, and disk driver circuits. The consumer electronic products can include, but are not limited to, a mobile phone, a telephone, a television, a computer monitor, a computer, a hand-held computer, a personal digital assistant (PDA), a microwave, a refrigerator, an automobile, a stereo system, a cassette recorder or player, a DVD player, a CD player, a VCR, an MP3 player, a radio, a camcorder, a camera, a digital camera, a portable memory chip, a washer, a dryer, a washer/dryer, a copier, a facsimile machine, a scanner, a multi-functional peripheral device, a wrist watch, a clock, etc. Further, the electronic devices can include unfinished products.

(92) Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” The word “coupled”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Likewise, the word “connected”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the above Detailed Description using the singular or plural number may also include the plural or singular number respectively. The word “or” in reference to a list of two or more items covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list.

(93) Moreover, conditional language used herein, such as, among others, “can,” “could,” “might,” “e.g.,” “for example,” “such as” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or states. Thus, such conditional language is not generally intended to imply that features, elements and/or states are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or states are included or are to be performed in any particular embodiment.

(94) The above detailed description is not intended to be exhaustive or to limit the disclosed embodiments to the precise form disclosed above. While specific embodiments are described above for illustrative purposes, various equivalent modifications are possible within the scope of this disclosure, as those skilled in the relevant art will recognize

(95) The teachings provided herein can be applied to other systems, not necessarily the system described above. The elements and acts of the various embodiments described above can be combined to provide further embodiments.

(96) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure.

Claims

1. An acoustic wave filter comprising: a plurality of series resonators; a plurality of shunt resonators disposed between the series resonators and a ground, at least one shunt resonator of the plurality of shunt resonators including a substrate and a pair of interdigital transducer (IDT) electrodes disposed on the substrate, each of the IDT electrodes including a bus bar and a plurality of fingers extending from the bus bar, fingers of one IDT electrode arranged interleaved with respective fingers of the other IDT electrode, the plurality of fingers of the pair of IDT electrodes including a first group of fingers located at a center region of the pair of IDT electrodes and a second group of fingers located at edge regions on both sides of the center region, the first group of fingers having an average pitch distance shorter than an average pitch distance of the second group of fingers such to improve a response at a rejection band of the acoustic wave filter, the at least one shunt resonator being configured to induce a longitudinal mode response formed by waves confined in a resonant cavity of the acoustic wave filter, such to increase a bandwidth of the rejection band of the acoustic wave filter; and a dielectric film formed to cover at least a part of the substrate and the pair of IDT electrodes.
2. The acoustic wave filter of claim 1 wherein a pitch distance of each finger from one IDT electrode is defined with respect to adjacent fingers from the other IDT electrode.
3. The acoustic wave filter of claim 1 wherein a width of the center region is determined based on a desired performance of the acoustic wave filter.
4. The acoustic wave filter of claim 1 wherein a direction of which the fingers of the pair of IDT electrodes are extended is perpendicular to a propagating direction of a main acoustic wave.
5. The acoustic wave filter of claim 1 wherein the at least one shunt resonator has a resonant frequency in a range of the rejection band of the acoustic wave filter.
6. The acoustic wave filter of claim 5 wherein a longitudinal mode response of the at least one shunt resonator is induced in vicinity of the resonant frequency of the at least one shunt resonator.
7. The acoustic wave filter of claim 1 further comprising a pair of reflectors arranged so as to interpose the pair of IDT electrodes therebetween.
8. The acoustic wave filter of claim 7 wherein each of the pair of reflectors includes a plurality of fingers having an average pitch distance longer than the average pitch distance of the second group of fingers of the pair of IDT electrodes.
9. The acoustic wave filter of claim 1 wherein each finger of the first group of fingers has different pitch distance depending on a required performance of the acoustic wave filter.
10. A radio frequency module comprising: a packaging board configured to receive a plurality of components; an acoustic wave filter implemented on the packaging board, the acoustic wave filter including a plurality of series resonators and a plurality of shunt resonators disposed between the series resonators and a ground, at least one shunt resonator of the plurality of shunt resonators including a substrate and a pair of IDT electrodes disposed on the substrate, each of the IDT electrodes including a bus bar and a plurality of fingers extending from the bus bar, fingers of one IDT electrode arranged interleaved with respective fingers of the other IDT electrode, the plurality of fingers of the pair of IDT electrodes including a first group of fingers located in a center region of the pair of IDT electrodes and a second group of fingers located in edge regions on both sides of the center region, the first group of fingers having an average pitch distance shorter than an average pitch distance of the second group of fingers such to improve a response at a rejection band of the acoustic wave filter, the at least one shunt resonator being configured to induce a longitudinal mode response formed by waves confined in a resonant cavity of the acoustic wave filter, such to increase a bandwidth of the rejection band of the acoustic wave filter; and a dielectric film formed to cover at least a part of the substrate and the pair of IDT electrodes.
11. The radio frequency module of claim 10 wherein the radio frequency module is a front-end

module.

12. The radio frequency module of claim 10 wherein a pitch distance of each finger from one IDT electrode is defined with respect to adjacent fingers from the other IDT electrode.
 13. The radio frequency module of claim 10 wherein a width of the center region is determined based on a desired performance of the acoustic wave filter.
 14. The radio frequency module of claim 10 wherein a direction of which the fingers of the pair of IDT electrodes are extended is perpendicular to a propagating direction of a main acoustic wave.
 15. The radio frequency module of claim 10 wherein the at least one shunt resonator has a resonant frequency in a range of the rejection band of the acoustic wave filter.
 16. The radio frequency module of claim 15 wherein a longitudinal mode response of the at least one shunt resonator is induced in vicinity of the resonant frequency of the at least one shunt resonator.
 17. The radio frequency module of claim 10 further comprising a pair of reflectors arranged so as to interpose the pair of IDT electrodes therebetween.
 18. The radio frequency module of claim 17 wherein each of the pair of reflectors includes a plurality of fingers having an average pitch distance longer than the average pitch distance of the second group of fingers of the pair of IDT electrodes.
 19. The radio frequency module of claim 10 wherein each finger of the first group of fingers has different pitch distance depending on a required performance of the acoustic wave filter.
 20. A mobile device comprising the radio frequency module of claim 10.
 21. A radio frequency module comprising: a packaging board configured to receive a plurality of components; an acoustic wave filter implemented on the packaging board, the acoustic wave filter including a plurality of series resonators and a plurality of shunt resonators disposed between the series resonators and a ground, at least one shunt resonator of the plurality of shunt resonators including a substrate and a pair of IDT electrodes disposed on the substrate, each of the IDT electrodes including a bus bar and a plurality of fingers extending from the bus bar, fingers of one IDT electrode arranged interleaved with respective fingers of the other IDT electrode, the plurality of fingers of the pair of IDT electrodes including a first group of fingers located in a center region of the pair of IDT electrodes and a second group of fingers located in edge regions on both sides of the center region, the first group of fingers having an average pitch distance shorter than an average pitch distance of the second group of fingers such to improve a response at a rejection band of the acoustic wave filter, the at least one shunt resonator having a resonant frequency in a range of the rejection band of the acoustic wave filter, the at least one shunt resonator configured to induce a longitudinal mode response in vicinity of the resonant frequency of the at least one shunt resonator; and a dielectric film formed to cover at least a part of the substrate and the pair of IDT electrodes.
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